

Home Assignment :Jira Integration

Open VS code

File -New Window

File-Open Folder

E:\Vidhya_Assignment link\August 23rd\rag-mongo-demo-v12\rag-mongo-demo-v9

Give the below Prompt in copilot inside VS code

Act as a Senior Architect

Currently the existing application has ingested the test cases and userstories into Mongo DB. Test cases file and userstories file were converted to JSON file in 'Convert to JSON' section in the application. The converted Json file checkbox was selected under 'Embeddings & Store' section in application and when 'Create Embeddings' button was clicked, the test cases and User stories were injected to Mongo DB. Test cases are getting generated for the user stories in the application. Now we need to integrate Jira with the existing application, fetch the user from Jira, Ingest into MongoDB, Retrieve existing context, generate test cases and display results.

[Mandatory] Integrate 'Jira' with existing application

[Mandatory] Fetch the 'User story' from Jira

[Mandatory] Ingest the fetched user story into 'MongoDB'

[Mandatory] Retrieve the existing/relevant context from MongoDB

[Mandatory] Use the Jira user story along with the retrieved context to generate test cases using the existing functionality

[Mandatory] Display the generated test cases in frontend

[Do not] Don't rebuild the existing ingestion,retrieval,or test case generation functionality.

[Strict] Focus on the Jira integration and connecting it with the existing flow

Convert to Json section in application:

The data type dropdown should be displayed with 'Jira' along with the existing drop-down values 'Test Cases' and 'User Stories'. On selecting the Jira story and clicking on 'Convert Jira' button, the selected user story in Jira should be converted to Json format.

Embeddings & Store section in application :

The converted Json file of Jira userstory should be listed. On selecting the checkbox against the Json file and clicking on 'Create Embeddings', the Jira story should be uploaded in the Mongo db.

Jira test case generation(Create this section newly in the application)'

The user stories available in Jira should be displayed. On selecting a user story and clicking on generate testcases, the test cases should be generated for the selected user story from Jira

Need to create test case to technically validate the functionality

Step 2 :

Create Jira API Key in Jira and make a note of it(Refer July 19th folder to refer steps)

Prompt 2 :

Plan mode :

Need to add "Connect Jira" button to establish the connection

Base URL :

<https://amvidhya1.atlassian.net/jira/software/projects/KAN/list?jql=project%20%3D%20KAN%20ORDER%20BY%20cf%5B10019%5D%20ASC>

Email id: amvidhya1@gmail.com

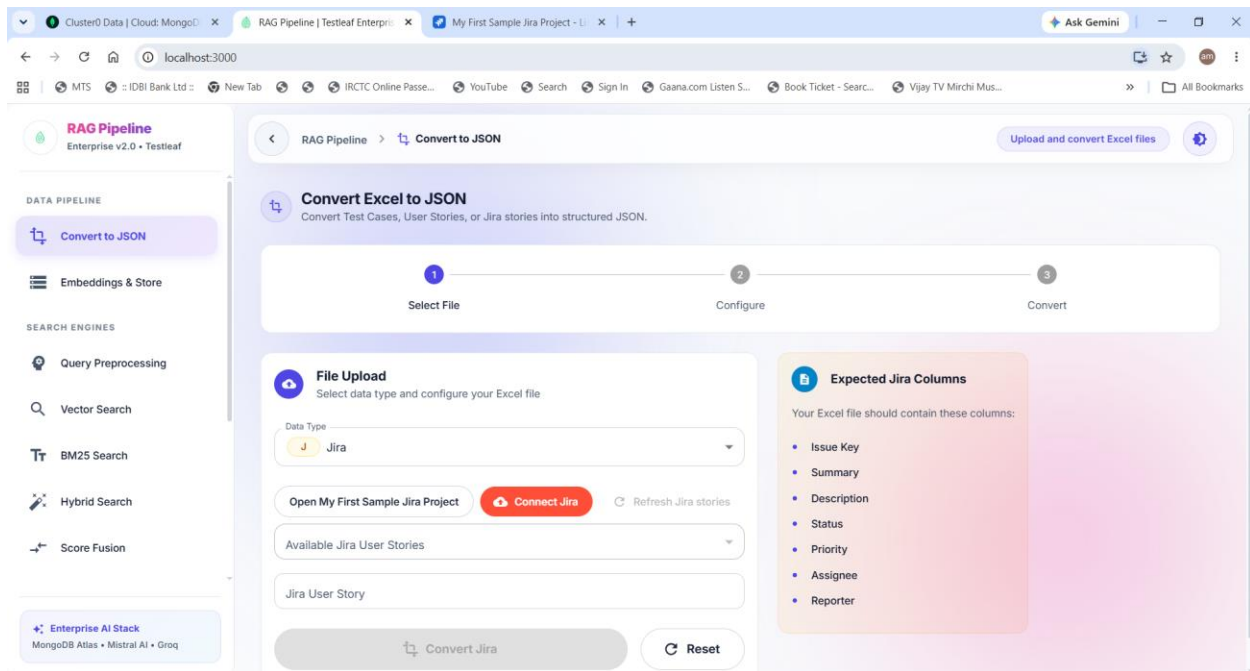
Jira API key :

ATATT3xFfGF09RgL7D1GT6PSP1gVblz1Z9hjHdrUidNJ_jGcRJ5N_b_VmME-
d74zvpd8lOjdwrtstmjxyBZM30OgNHdJt5CKjP1Z4JtSALg1OaAEGEKtLrFVVseeqqNGNlgzuc
PZY0oiJcyA94XfE3Zz9Hc2I50zQT9cn8twEWmOf0TSiwxoGPY=AB8A0E7C

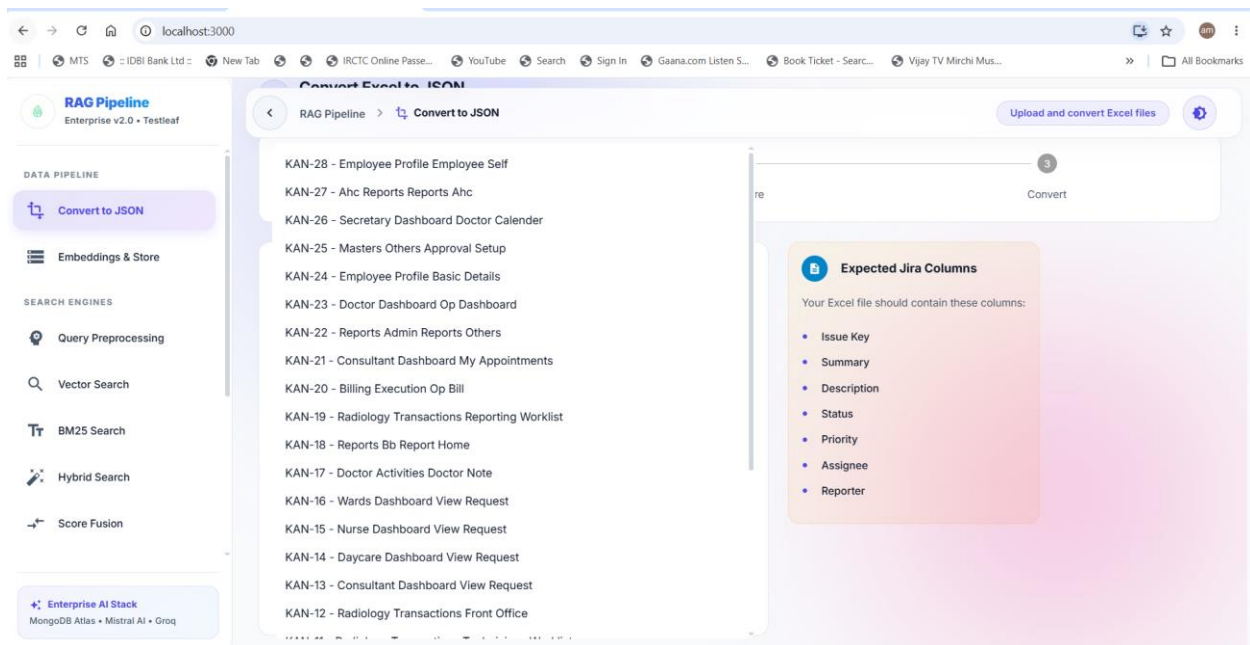
Add the variables in .env file

Give the below prompt in agent mode

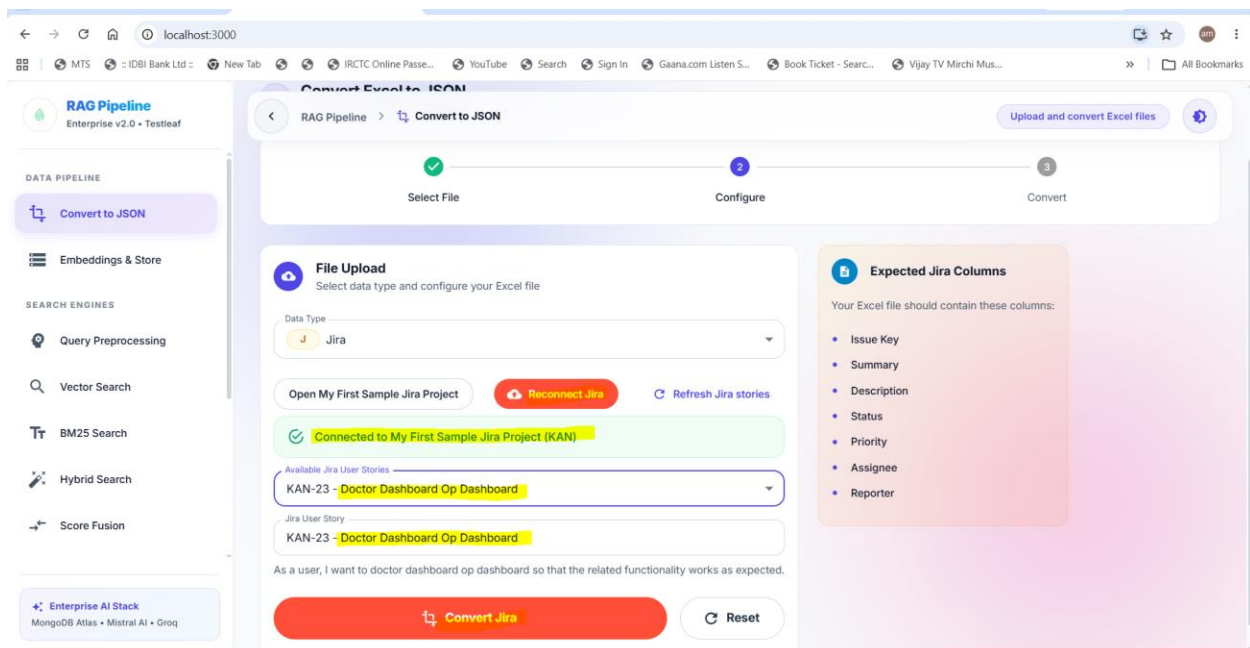
Go ahead with the implementation



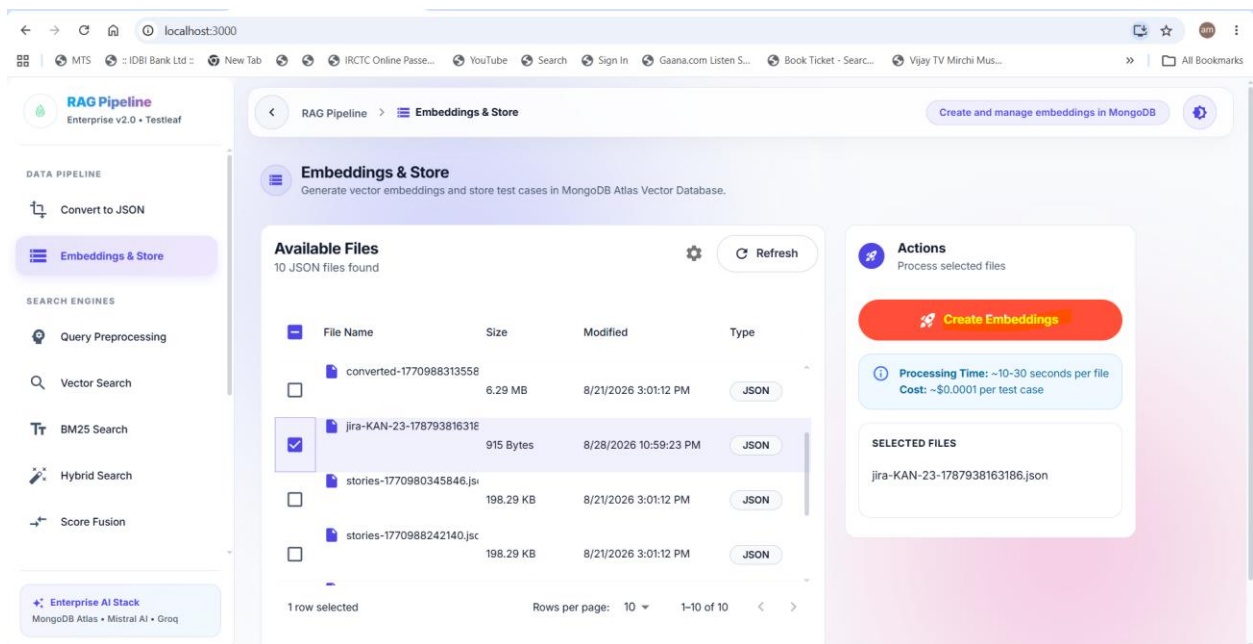
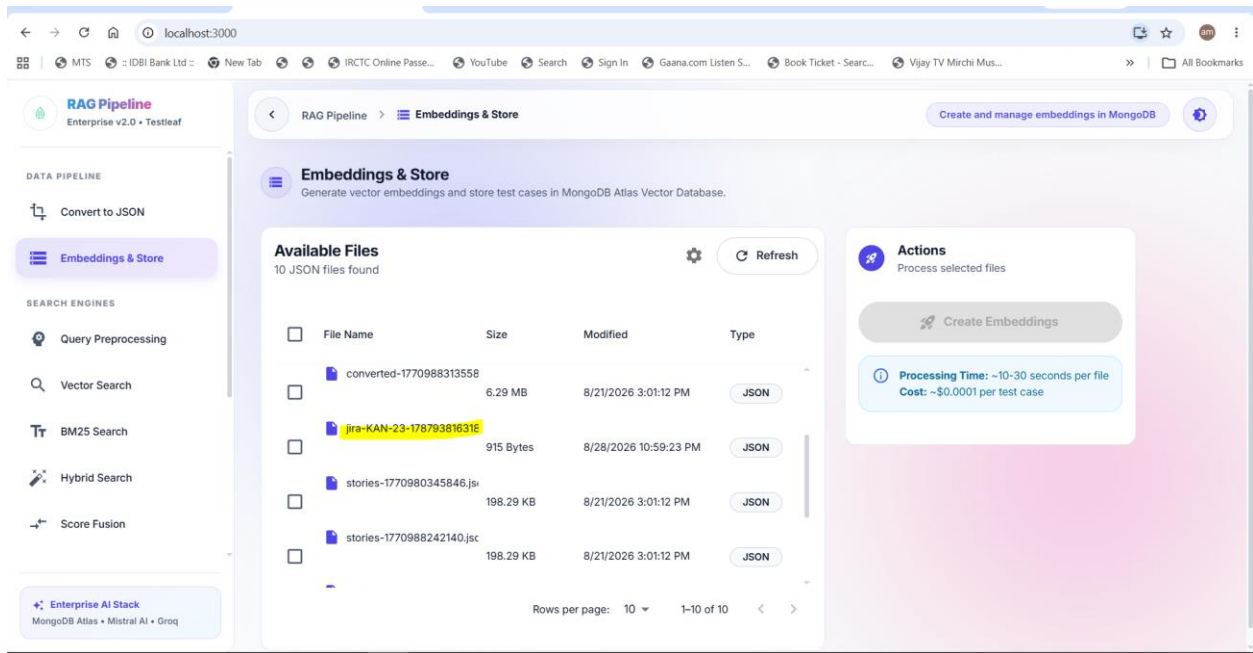
Click on 'Connect to Jira'. Select a user story from the drop-down option displayed with Jira user stories.



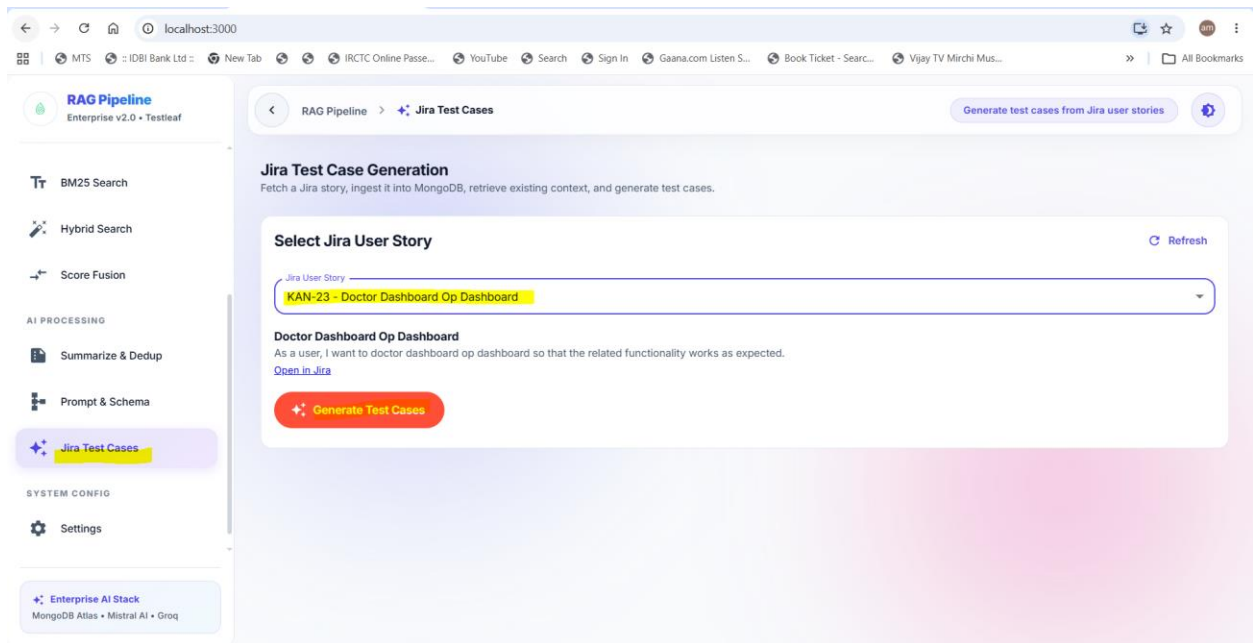
Select one Jira User story and click on 'Convert Jira'. Conversion will be successful.



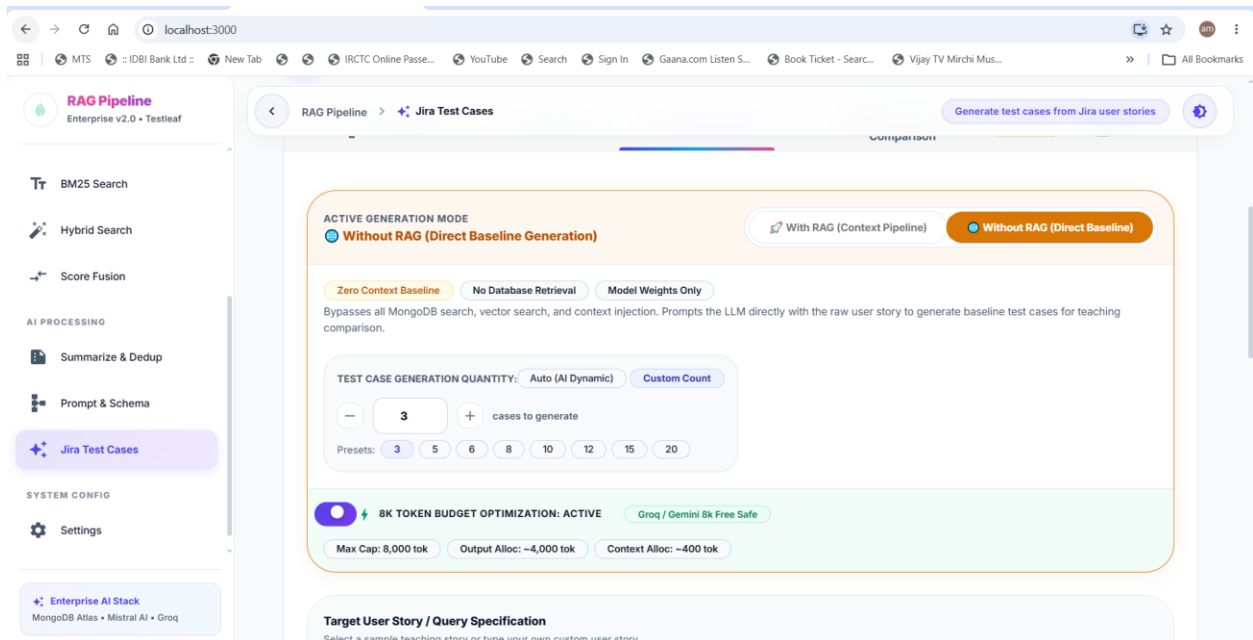
Select the checkbox against the converted Json Jira user story



Navigate to 'Jira Test Cases' and select the Jira user story and click on Generate test cases.



Select the number of test cases the need to be generated



Click on Generate Test cases. Test cases will be generated for the selected Jira user story

localhost:3000

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Go to Comparison Tab

Save Config

Generate test cases from Jira user stories

RAG Pipeline > Jira Test Cases

BM2S Search

Hybrid Search

Score Fusion

AI PROCESSING

Summarize & Dedup

Prompt & Schema

Jira Test Cases

SYSTEM CONFIG

Settings

Enterprise AI Stack

MongoDB Atlas • Mistral AI • Groq

Generated Test Cases Without RAG (Direct Baseline)

Mode: Without RAG Model: openai/gpt-oss-120b 0 DB Retrieval (Zero Context) 2,956 / 8000 Tok (37% Budget Safe) Export CSV

ID	Title	Module	Priority	Type	Preconditions	Test Steps	Expected Results
TC_0002	Negative Flow – Unauthorized Access and Session Expiration	Doctor Dashboard	P1	Functional	Nurse user account exists with role=Nurse. System idle timeout is configured to 15 minutes. No active session for the nurse.	1. Navigate to the hospital portal login page. 2. Log in using the nurse credentials. 3. Attempt to navigate directly to the URL **/doctor/dashboard** via browser address bar. 4. Observe the system response. 5. Log out of the nurse account. 6. Log in again using a valid doctor account. 7. Remain idle on the dashboard for 16 minutes without any interaction. 8. Attempt to perform any action (e.g., click **Refresh** button) after the idle period.	user remains on an error page. • Step 8: Session is terminated; user is redirected to login page with message "Session expired due to inactivity". No residual session cookies remain. • Audit log records two events: unauthorized access attempt (user

localhost:3000

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ID	Title	Module	Priority	Type	Preconditions	Test Steps	Expected Results
TC_0003	Boundary & Performance – Dashboard Rendering with Maximum Patient Load	Doctor Dashboard	P2	Integration	Doctor account exists with 5,000 patients linked (UHID range 100000-105000). Test environment configured with performance monitoring tools (e.g., JMeter, Lighthouse).	4. Apply a filter to show patients admitted in the last 24 hours (date range edge case). 5. Verify that the filtered result set returns within 3 seconds. 6. Scroll to the bottom of the patient list to trigger lazy-loading of additional rows. 7. Observe memory usage on the client (browser dev tools) and ensure it does not exceed 250MB. 8. Export the patient list to CSV and confirm the file contains exactly the number of rows displayed.	Step 7: Browser memory consumption remains ≤ 250MB throughout the session. • Step 8: Exported CSV file contains the same row count as displayed (validation of data integrity) and includes encrypted PHI fields that are decrypted